Time-Series Autoencoder - High-Level Project Steps

Flowchart summary of the notebook from raw data to suspicious window detection

1. Load raw PaySim data

Read transaction data and keep relevant columns such as step, type, amount, balances, and fraud label.

2. Convert transactions into time-step KPIs

Aggregate by step to create one row per time period with volume, amount, fraud count, fraud rate, and transaction-type shares.

3. Split by time order

Create train, validation, and test periods without shuffling so the model learns from earlier time and is tested on later time.

4. Preprocess features

Scale numeric KPI columns and keep the feature matrix ready for sequence learning.

5. Create sliding windows

Build fixed-length sequences so the model learns patterns across multiple consecutive time steps, not one step alone.

6. Train the LSTM autoencoder

Input sequence and target sequence are the same. The model learns to reconstruct normal time behaviour.

7. Calculate reconstruction error

Measure how different the reconstructed sequence is from the original sequence.

8. Set anomaly threshold

Use validation reconstruction error and choose a percentile threshold to define suspicious windows.

9. Flag suspicious windows

Any test window with error above the threshold is marked as suspicious.

10. Interpret with BI charts

Compare error over time, suspicious windows, fraud count, KPI summary, and business meaning.

Final business outcome

The notebook does not directly predict fraud labels. It learns normal time behaviour and highlights unusual time windows for investigation. Fraud counts are used later only to interpret whether suspicious windows align with risky periods.